UNIT 4 · RAPID REVISION

Formula and Fact Sheet

Key constants and formulas

Item	Value
γ for air	1.4 (cp = 1.005, cv = 0.718 kJ/kg·K)
R for air	0.287 kJ/kg·K
Carnot efficiency	η = 1 − T_low/T_high (900 K / 300 K → 66.7%)
Otto efficiency	η = 1 − 1/r^(γ−1); r = 8 → 56.5%
Torque from power	T (N·m) = 9550 × P (kW)/N (rpm)
Indicated power	IP = p_m·L·A·n·K/60,000 (n = N/2 for 4-stroke)
Mean piston speed	2LN/60 m/s (tractor diesels ~6–12 m/s)
Torque backup	(T_max − T_rated)/T_rated × 100 — 15–30% = good lugging
Firing interval (4-stroke)	720°/K: 3-cyl 240°, 4-cyl 180°, 6-cyl 120°
Cooling load	Q = ṁ·c_p·ΔT
Flywheel inertia	I = ΔE/(C_s·ω²)
Clutch torque capacity	T = n·μ·W·R_m; R_m = (Ro + Ri)/2
Differential speeds	N_L + N_R = 2·N_crown
Weight transfer under pull	ΔR = P·h/L
Ackermann condition	cot δo − cot δi = track/wheelbase

Standards, settings and limits

Item	Value
PTO 540 rpm shaft	35 mm, 6 splines (IS 4931 / ISO 500)
PTO 1000 rpm shaft	21 splines
Cat-I three-point linkage	Lower pin ~22 mm, top ~19 mm, mast 460 mm
Cat-II three-point linkage	Lower pin ~28 mm, top ~25.5 mm, mast 610 mm
BS-VI diesel	Sulphur ≤ 10 ppm; cetane ≥ 51
TREM IV	January 2023, >50 hp tractors
TREM V	1 Oct 2026 for 56–560 kW & 8–19 kW; PM 0.015 g/kWh
E20 petrol	Mandatory nationwide from 1 April 2026
Nebraska test law	1919; maximum pull reported at 15% slip
CFMTTI Budni	Est. 1955 — India's central tractor-testing station
Valve clearance (cold)	~0.25 mm intake / 0.30 mm exhaust
Injector opening pressure	~170–250 bar (conventional)
Thermostat	Begins opening ~80–85 °C; engine runs 80–95 °C
Engine oil pressure	~2–4 bar at rated speed
Clutch free pedal play	~20–25 mm
Drawbar height	~330–500 mm

Maintenance intervals (typical)

Interval	Jobs
Daily (10 h)	Fuel, oil level, radiator water, air cleaner, tyres, leaks
50 h	Greasing, battery electrolyte, fan-belt tension, wheel nuts, oil-bath cleaner oil
250 h	Engine oil + filter, fuel filter, valve clearance, tyre pressure/rotation
500 h	Injectors, transmission/hydraulic oil levels, steering & brakes
1000 h	Transmission-hydraulic oil & filters, FIP calibration, alternator, full service

Thermodynamics, cycles and fuels (Topics 1, 7a)

- γ_air = 1.4; R = 0.287 kJ/kg·K; η_carnot = 1 - T2/T1; η_otto = 1 - 1/r^(γ-1) (r = 8 → 56.5%); Diesel adds heat at constant p; dual = both.
- Same r: Otto > dual > Diesel; same maximum pressure: Diesel best; SI r 6–10 (knock/octane), CI r 14–22 (cetane).
- HSD: cetane ≥ 51, S ≤ 10 ppm (BS-VI), CV ≈ 42–45 MJ/kg; octane ↑ ⇒ cetane ↓; CI combustion: delay → premixed → diffusion.
- E20 pan-India from 1 Apr 2026 (20% reached Nov 2025); TREM IV Jan 2023 (>50 hp); TREM V from 1 Oct 2026 (56–560 kW & 8–19 kW), PM 0.015 g/kWh, EU Stage V aligned.

Engine systems (Topic 2)

- Camshaft = ½ crank speed; valve overlap at TDC; clearance ~0.25/0.30 mm — too small ⇒ burnt valves, too large ⇒ noise and power loss.
- FIP: inline (helix + rack) vs distributor; injector 170–250 bar; bleed after a filter change; oil-bath vs dry air cleaner; turbo +20–30%.
- CI = quality governing; SI = quantity; variable-speed flyweight governor; droop and hunting.
- Thermostat ~82 °C; run 80–95 °C; pressure cap raises the boiling point; overcooling causes dilution and wear.
- Full-pressure lubrication; oil pressure 2–4 bar; SAE = viscosity, API = service class; blue smoke = oil, black = fuel/air, white = misfire/water.

Engine calculations (Topics 3–4)

- T = 9550·P/N; IP = p_m·L·A·n·K/60,000 (n = N/2, 4-stroke); mean piston speed = 2LN/60; torque backup 15–30% = lugging ability.
- Firing interval = 720/K (4-stroke): 3-cyl 240°, 4-cyl 180°, 6-cyl 120°; orders 1-3-4-2 (4-cyl), 1-5-3-6-2-4 (6-cyl); cranks 1 & 4 / 2 & 3 paired.
- Heat balance in thirds: ~⅓ BP, ~⅓ cooling, ~⅓ exhaust (+ small friction); Q = ṁcΔT for the radiator load.
- Side thrust = F·tanφ (piston slap); T = F_t × r peaks ~30° ATDC; I = ΔE/(C_s·ω²); single-cylinder → largest flywheel.

Transmission, PTO and drawbar (Topics 5–6)

- Flow: engine → clutch → gearbox → differential → final drive → wheels; overall η ≈ 88–92%; PTO ≈ 87–92% of engine power.
- Clutch T = n·μ·W·R_m (n = 2 for single plate); free play 20–25 mm; dual clutch = live PTO; capacity 1.25–1.5× engine torque.
- Synchromesh = speed-matching cones; 8F + 2R typical; v = πDN_wheel; wheel torque = T_e × reduction × η.
- Diff: N_L + N_R = 2N_crown; equal torque split → diff lock for one-wheel slip; final drive = last reduction at the wheel + clearance; steering brakes on bull-pinion shafts.
- PTO: 540 rpm/6-spline/35 mm; 1000 rpm/21-spline (IS 4931/ISO 500); transmission vs live vs independent; drawbar height ~330–500 mm, swinging type for trailed implements.

Chassis, hitching, hydraulics and steering (Topics 8–10)

- R_f = W·x/L; 2WD static ≈ 35/65 front/rear; ΔR = P·h/L; rearward overturn: high hitch + torque reaction; keep ≥ 20% weight on the front; side slope tan θ = B/2h.
- 3-point: 2 lower + 1 top link; Cat-I mast 460 mm/pins ~22 mm; Cat-II 610 mm/28 mm; virtual hitch point = link convergence; mounted > semi-mounted > trailed for weight transfer.
- Position control = constant height (linkage feedback); draft control = constant draft via top-link/lower-link sensing → auto depth + weight transfer; mixed control for variable soils.
- Ackermann: cot δo - cot δi = track/wheelbase; inner angle larger; toe-in 3–12 mm; caster = self-centring; hydrostatic (Orbitrol) power steering is standard now.

Tests, emissions, trends and maintenance (Topics 7, 11–12)

- Most useful criterion = maximum drawbar power (PTO power for PTO implements) — Liljedahl p. 403; sound in dB(A) since 1971; Nebraska law 1919; manufacturer selects the test tractor; 1986: OECD-station/SAE-ASAE codes accepted; results published; maximum pull at 15% slip.
- Power corrected to standard atmosphere; OECD codes = international benchmark (NIAE Silsoe); India = CFMTTI Budni (1955), BIS codes, subsidy-linked.
- NOx ↔ PM trade-off; EGR cuts NOx, DPF traps PM, DOC burns HC/CO, SCR uses DEF.
- Trends: CRDI, CVT, e-tractors (PRIMA ET11), auto-steer, telematics. Maintenance: air cleaner #1 in dust; oil/filter 250 h; transmission-hydraulic 1000 h; power tiller 7–12 hp, rotary-tiller based.

Further reading

- Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units (Ch. 14, Tractor Tests and Performance).
- Kepner, Bainer & Barger — Principles of Farm Machinery (hitching); Jagdishwar Sahay — Elements of Agricultural Engineering.
- BIS: IS 4931 (PTO), IS 4468 (three-point linkage); ISO 500, ISO 730; OECD tractor test codes; SAE J708 / ASAE S209.
- MoRTH/CMVR TREM notifications (egazette.gov.in); MoPNG/PIB releases on ethanol blending.
- CFMTTI Budni test procedures; NPTEL lectures on farm power and IC engines.